DATA ANALYTICS PROJECT

BANKING – LOAN POLICY & RISK INTELLIGENCE

A comprehensive analytics solution integrating Microsoft Excel, Power BI and Python to deliver actionable insights into Instagram content performance, audience engagement and content trends.

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Course	Data Analytics Certification (180 Hours)
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Abstract

The banking sector relies on effective loan assessment and risk management to maintain financial stability, reduce lending risks, and make informed lending decisions. Evaluating loan applications requires the analysis of several factors, including applicant income, credit score, existing financial obligations, employment stability, requested loan amount, and repayment capacity. Without structured analysis, it can be difficult to identify high-risk applications, evaluate lending eligibility, and determine whether requested loan amounts comply with established lending policies.

The project, “Banking – Loan Policy & Risk Intelligence,” focuses on analysing loan application data and developing an interactive business intelligence solution using Microsoft Power BI. The project aims to transform applicant-level data into meaningful insights that support loan eligibility assessment, policy compliance monitoring, and risk analysis.

The dataset contains applicant information and financial attributes such as age, gender, income, loan amount, credit score, employment type, employment years, existing loan count, existing EMI, debt-to-income percentage, loan type, and original loan status. Additional policy-related fields, where available in the prepared dataset, support the comparison of requested loan amounts with policy limits and the classification of loan eligibility and risk categories.

The project involves examining the dataset, preparing the data for analysis, creating calculated fields and measures, and developing interactive visualizations in Power BI. The dashboard enables users to explore loan applications, compare loan products, examine eligibility patterns, evaluate applicant risk, and assess the relationship between requested loan amounts and policy maximums.

Python and its data analysis libraries are also used to generate supporting visualizations from the banking dataset. These visualizations complement the Power BI dashboard by providing additional views of the financial data and supporting the interpretation of loan policy and risk-related patterns.

The completed analytical solution provides a structured approach to understanding loan applications and monitoring lending policy indicators. The findings can help banking professionals identify areas requiring further review, improve the consistency of loan assessment, and support data-driven lending decisions. The project demonstrates how business intelligence and data visualization can convert financial data into practical information for loan policy evaluation and risk monitoring.

1. Introduction

1.1 Project Overview

Banking institutions evaluate loan applications based on factors such as income, credit score, employment stability, and existing financial commitments. The Banking – Loan Policy & Risk Intelligence project analyses these factors to understand loan eligibility and lending risk. Microsoft Power BI is used to develop an interactive dashboard displaying key financial indicators and comparing requested loan amounts with policy-defined maximum limits. Python libraries such as Pandas, NumPy, and Matplotlib support data analysis and visualization. Overall, the project demonstrates data preparation, financial analysis, and interactive reporting to support loan policy evaluation and risk assessment.

1.2 Problem Statement

Loan assessment requires analysing multiple factors, including applicant income, credit score, employment stability, existing financial commitments, and requested loan amounts. Without a structured analytical system, comparing applications, monitoring lending indicators, and identifying potential policy-limit issues can be challenging. This project addresses these challenges by developing an interactive Power BI dashboard to analyse loan applications, compare requested loan amounts with policy limits, and visualize applicant characteristics and risk-related indicators, supporting informed analytical decision-making.

1.3 Dataset

The project uses an Excel dataset named “Banking_Loan_Policy_PowerBI_Ready.xlsx”, containing 1,000 records of banking loan application data covering the period 2024–2025. The dataset includes applicant details, financial information, employment attributes, and loan characteristics such as Applicant_ID, Income, LoanAmount, CreditScore, Loan_Type, Employment_Years, Existing_EMI, and DTI_Percent. Additional fields, including Policy_Max_Loan, Eligible Loan Amount, Eligibility_Status, and Risk_Category, support loan eligibility analysis, policy-limit comparisons, and risk classification. The dataset is analysed using Microsoft Power BI and Python to identify lending patterns and support loan policy assessment.

1.4 Tools Used

Tool / Technology	Purpose
Microsoft Excel	Source data storage and dataset input
Python	Data preparation, time series analysis and forecasting
Pandas	Data loading, cleaning, grouping
NumPy	Numerical data handling
Matplotlib	Static data visualization
Statsmodels	Exponential Smoothing time series forecasting
Power BI	Interactive dashboard and visual exploration

2. Data Preparation — Excel & Power Query

2.1 Data Loading

The banking loan dataset, Banking_Loan_Policy_PowerBI_Ready.xlsx, was imported into Power BI Desktop using the Get Data option. The data was inspected in Power Query Editor to review column names, data types, and potential data-quality issues. This preparation ensured that the dataset was organized and ready for analysis, calculations, and dashboard development.

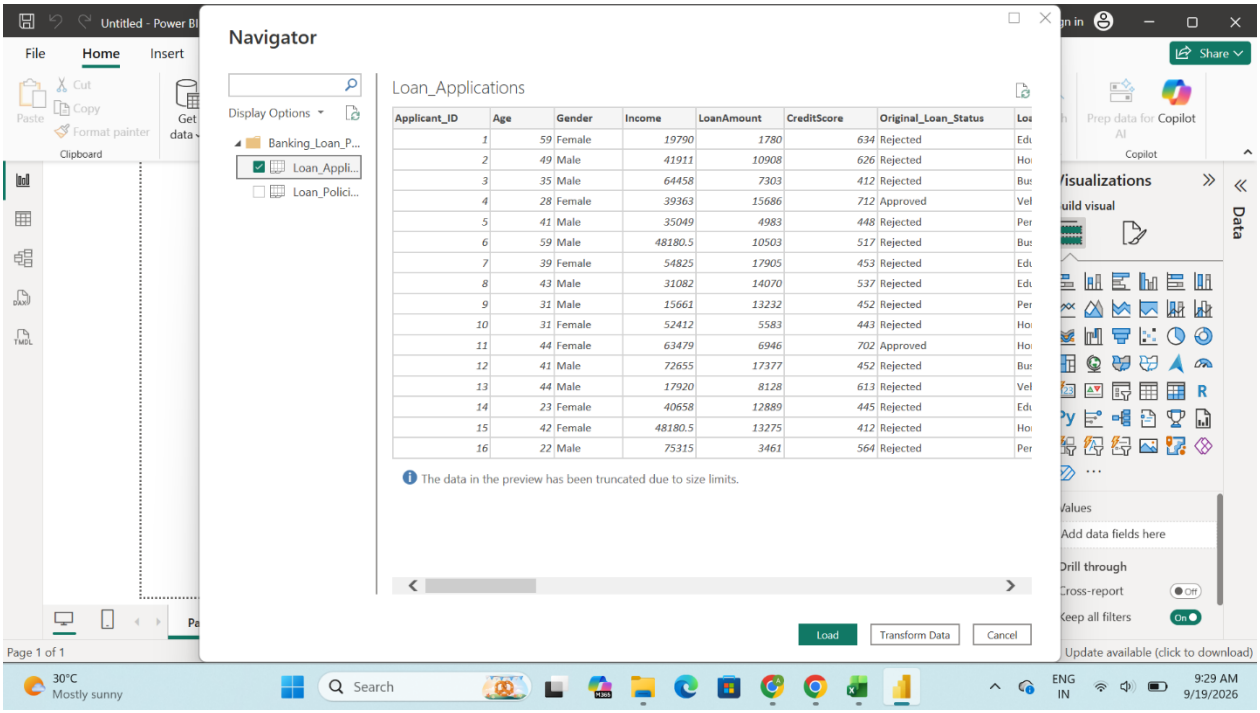


Figure 1 — *Raw data loaded into Power Query before transformation*

2.2 Data Type Corrections

Data types were reviewed and corrected in Power BI Power Query Editor to ensure accurate data analysis and visualization. Applicant identifiers and categorical fields, such as Gender, Loan_Type, and Employment_Type, were assigned appropriate data types. Financial fields, including Income, LoanAmount, Existing_EMI, and Policy_Max_Loan, were set to suitable numeric types, while Age, CreditScore, and Existing_Loan_Count were treated as whole numbers where appropriate. Date fields, such as Application_Date, were assigned a date format where available. Eligibility and risk classification fields were treated as categorical data to support effective filtering and comparison. These corrections help prevent calculation errors, ensure consistent data aggregation, and improve the accuracy of dashboard visualizations and analytical results.

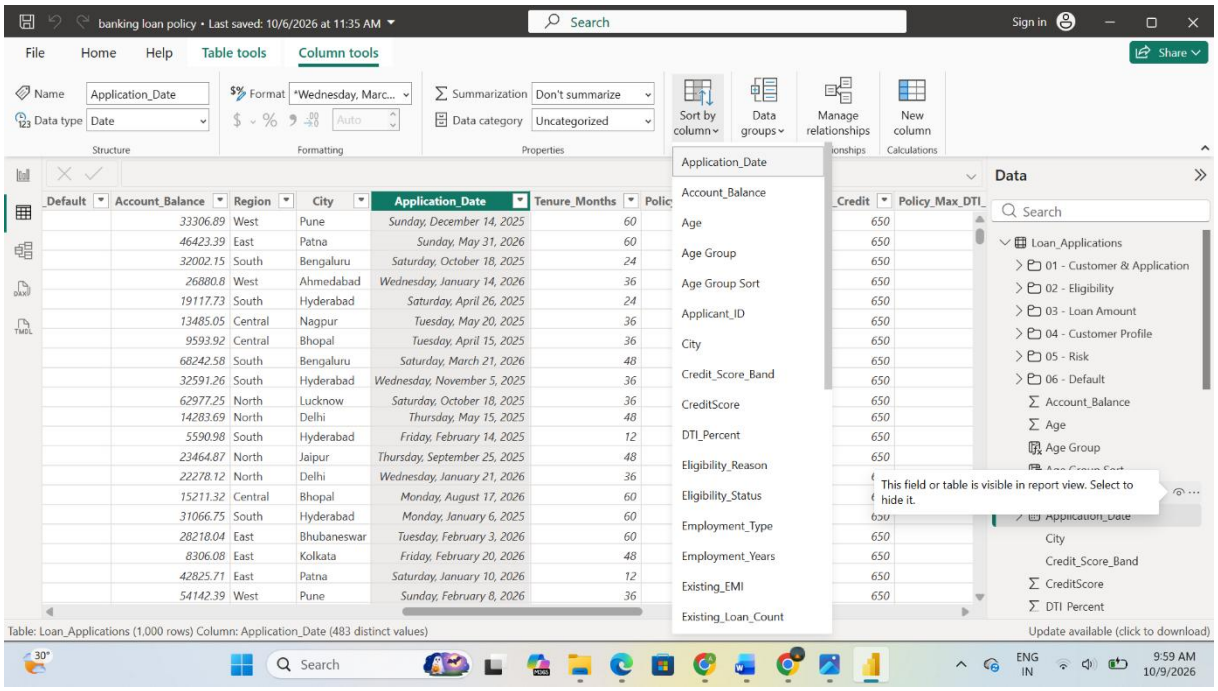


Figure 2 — *Data type conversion into datetime — post_datetime*

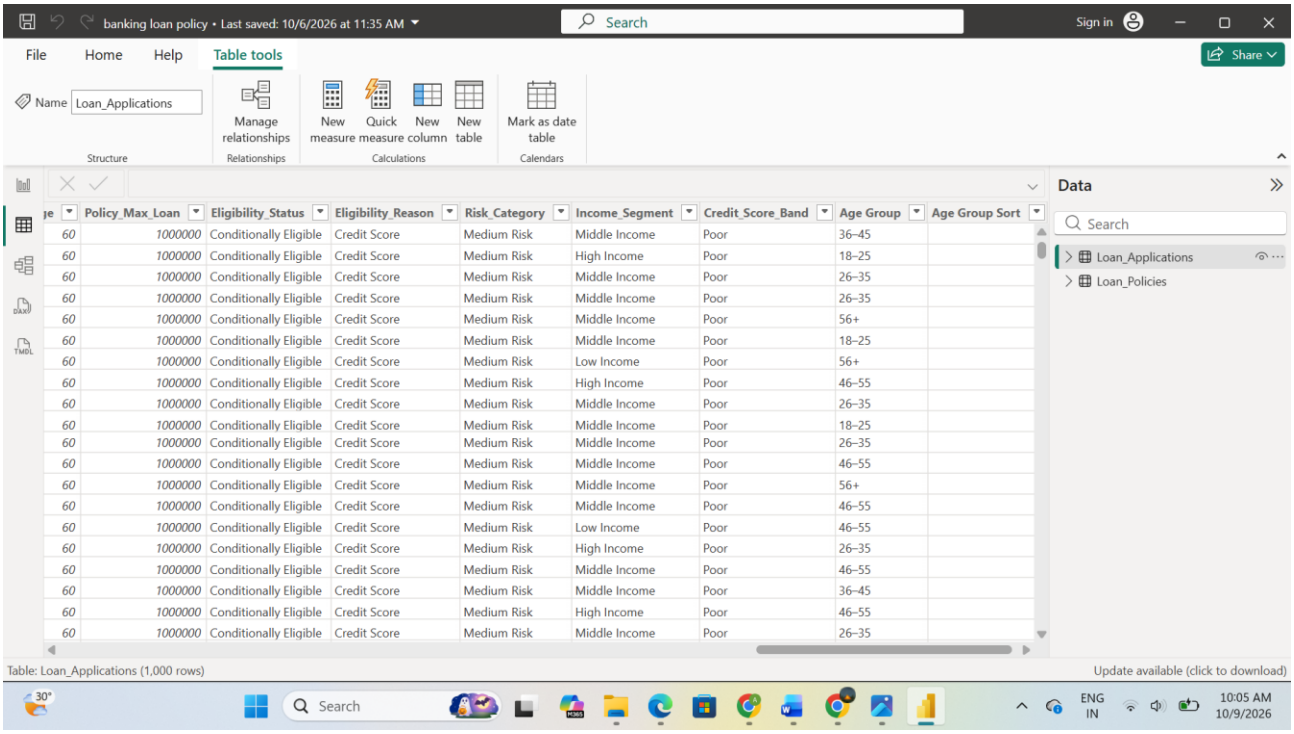


Figure 3 — *Data type correction for applicant and categorical fields*

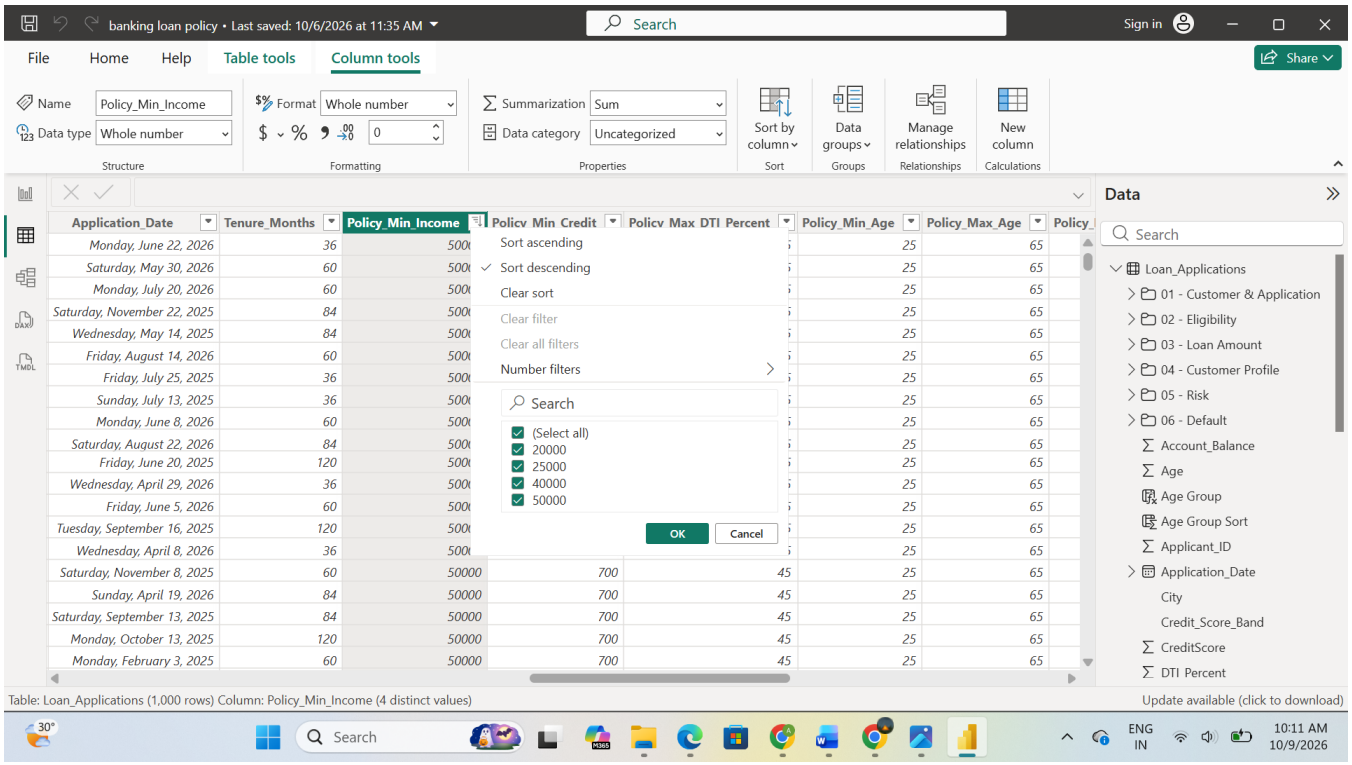


Figure 4 — *Numeric data type correction for financial attributes*

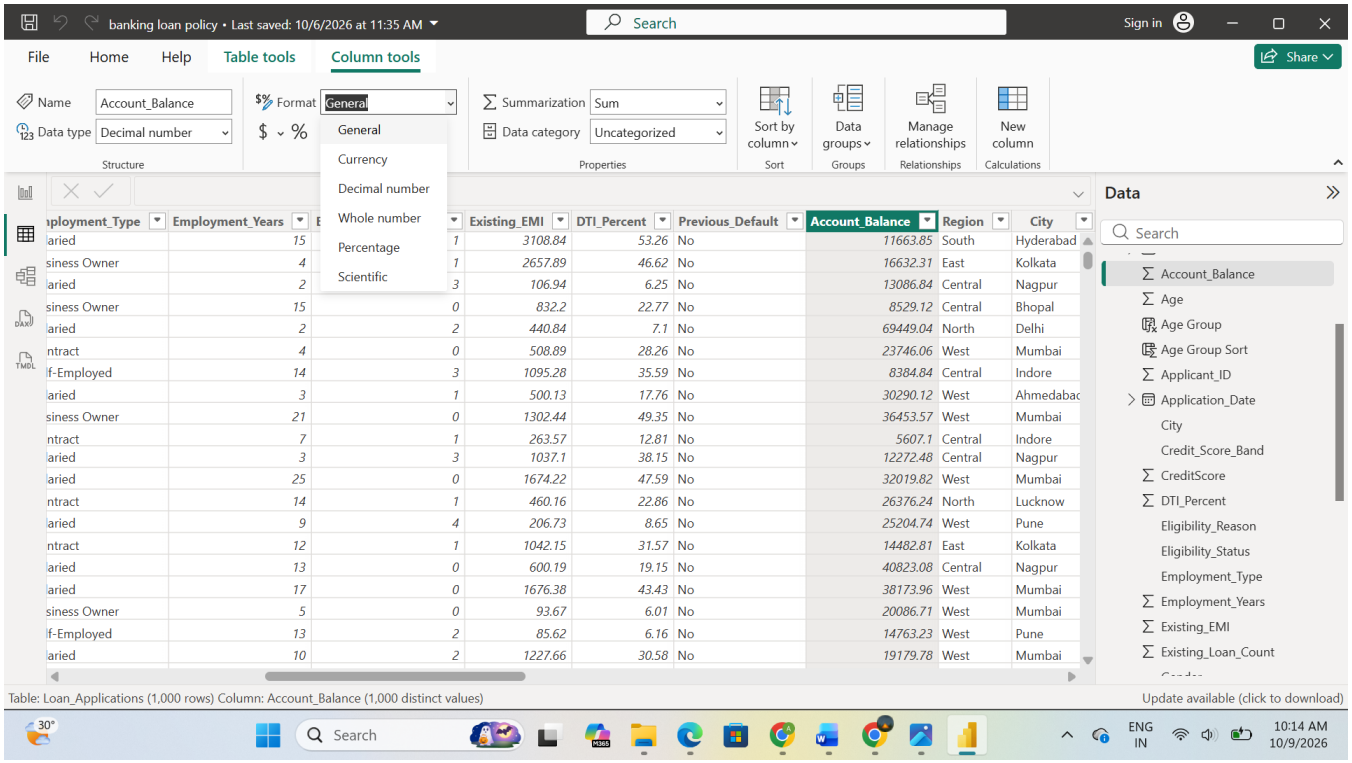


Figure 5 — *Data type conversion to decimal number*

2.3 Duplicate Detection and Removal

A Duplicate records were checked in Power BI Power Query Editor to maintain data accuracy and prevent incorrect application counts and financial totals. The Applicant_ID field was reviewed for repeated identifiers, and a full-row duplicate check was considered to identify identical records. Confirmed

duplicate records, if present, can be removed using the Remove Duplicates option. The row count before and after removal should be compared to validate the changes. This process helps maintain data consistency and ensures reliable calculations and dashboard reporting.

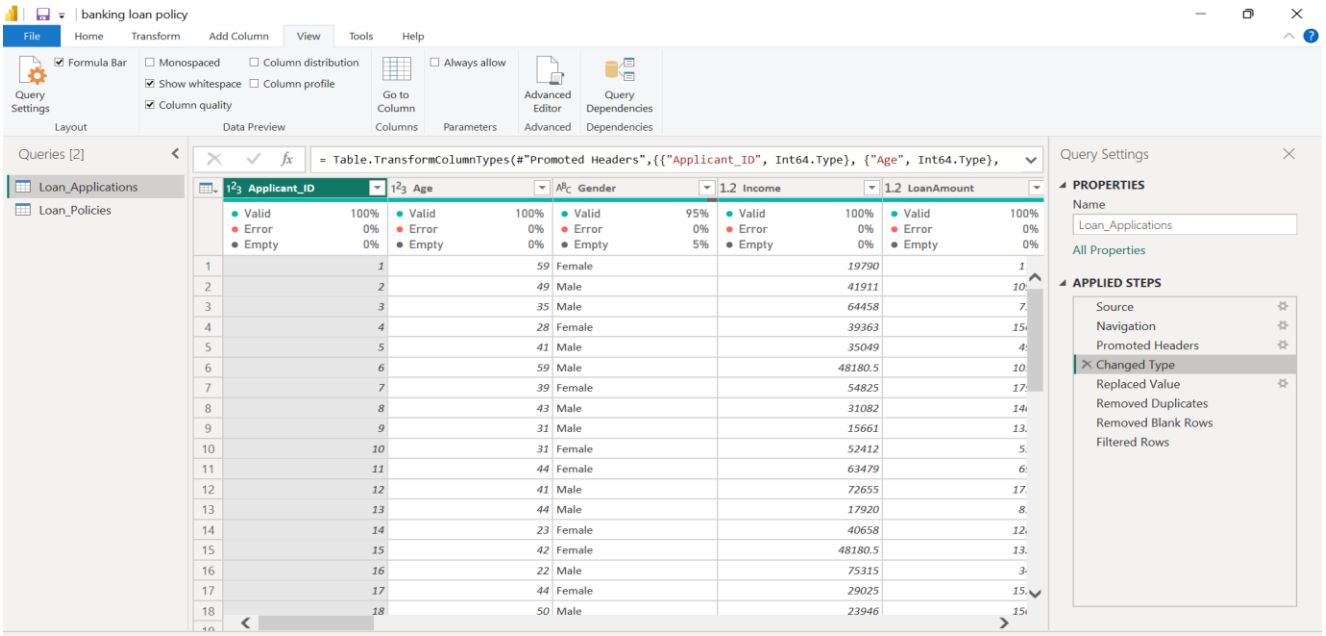


Figure 6 — *Row count before duplicate removal — Power Query status bar*

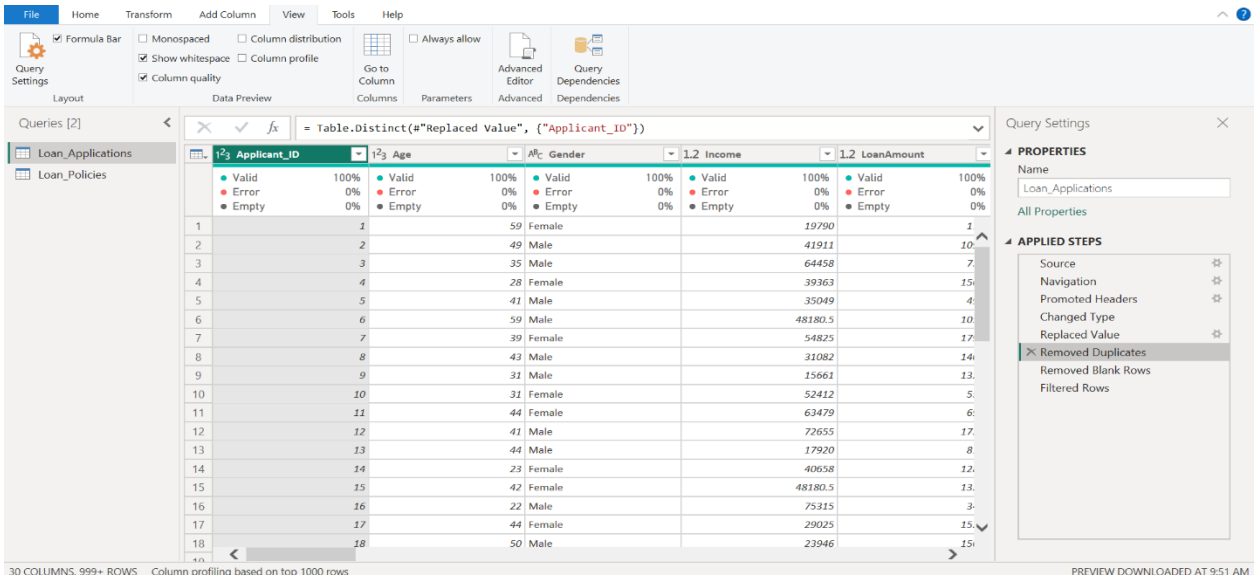


Figure 7 — *Row count after duplicate removal — Power Query status bar*

2.4 Column Removal

The essential columns were retained to support loan analysis and dashboard reporting. Fields such as Loan_Type, LoanAmount, Policy_Max_Loan, Application_Date, and Eligibility_Status enable loan category comparisons, policy-limit analysis, monthly trend analysis, and eligibility assessment.

2.5 Derived Column Creation

The Application_Date field supports monthly loan application trend analysis by organizing records chronologically. Date fields such as Year and Month may be created in Power BI for time-based reporting.

The eligibility rate is calculated by dividing eligible applications by the total number of applications. These fields and calculations help compare loan categories and analyse application trends.

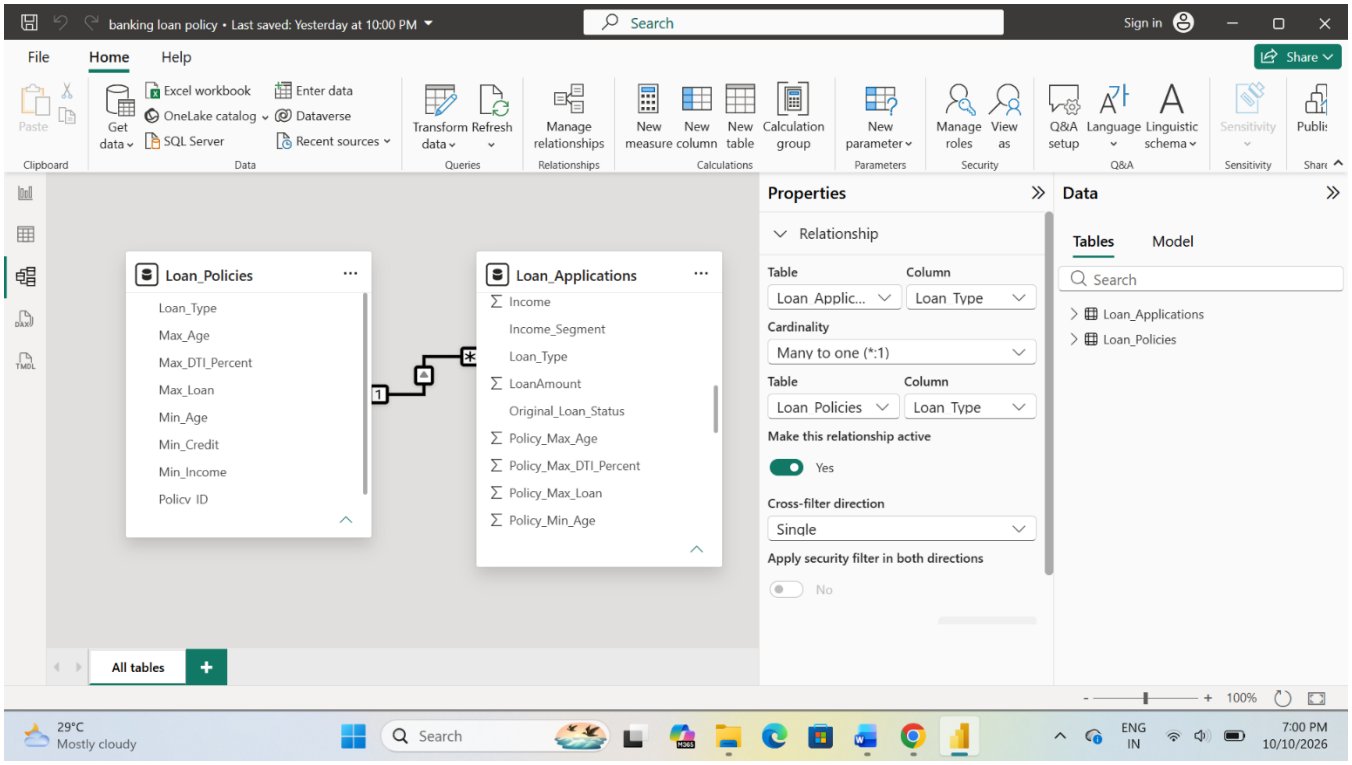


Figure 8 — *Date-related column or date grouping used for monthly analysis*

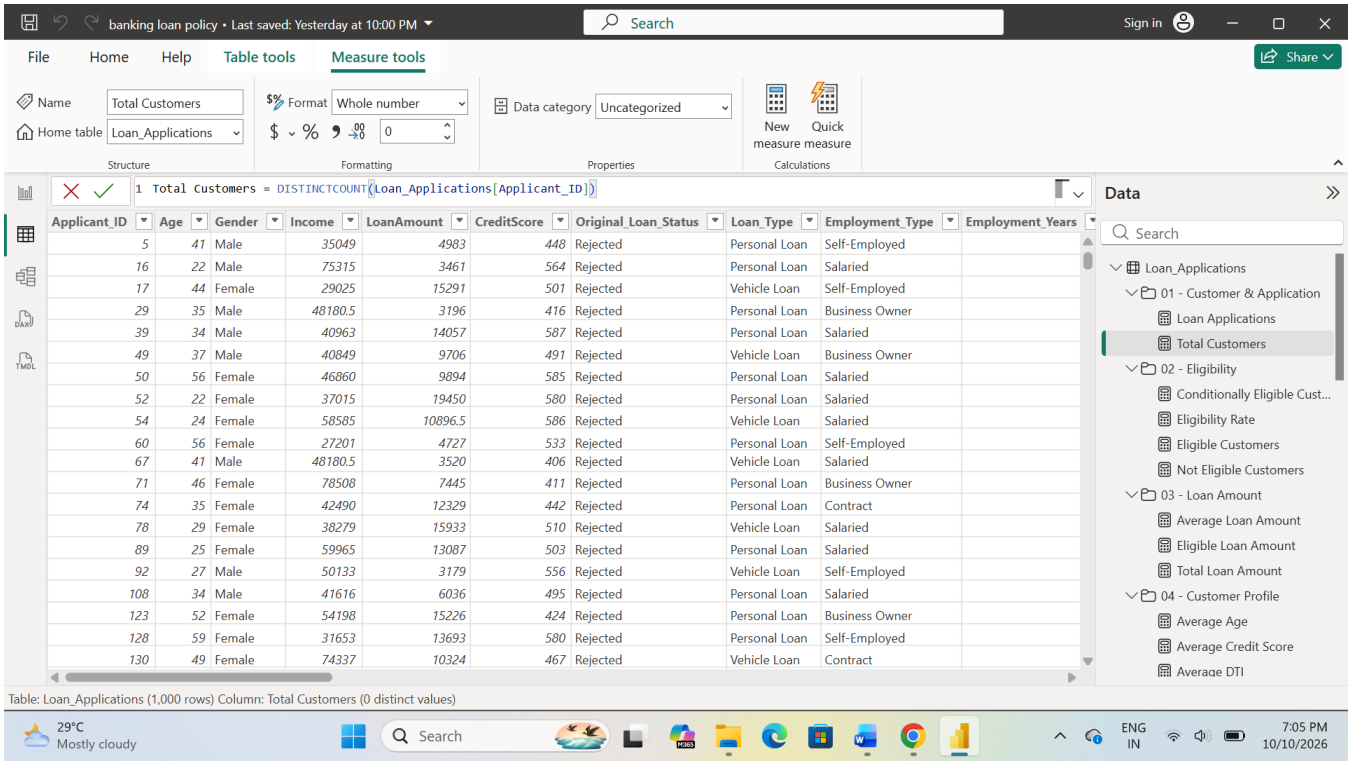


Figure 9 — *Eligibility calculation or measure used in the dashboard*

3. Power BI Dashboard

3.1 Data Model

The Power BI data model organizes the loan application dataset for dashboard calculations and visualizations. The principal fields used in the project include Loan_Type, LoanAmount, Policy_Max_Loan, Application_Date, and Eligibility_Status, which support the analysis of loan application distribution, product demand, monthly trends, eligibility rates, and policy maximums. Application_Date enables time-based grouping, while Loan_Type and Eligibility_Status provide categorical dimensions for comparison. The Model View can be used to inspect the table structure and any relationships implemented in the Power BI file.

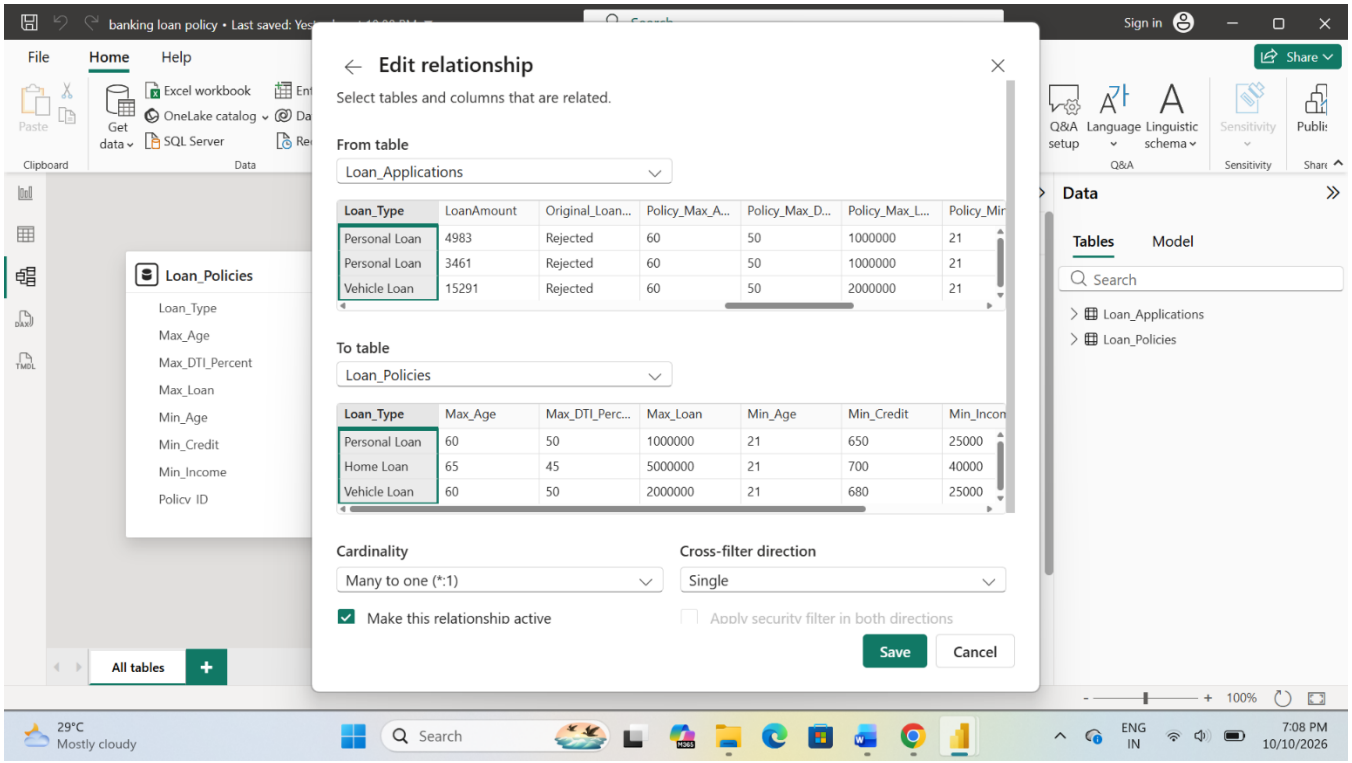
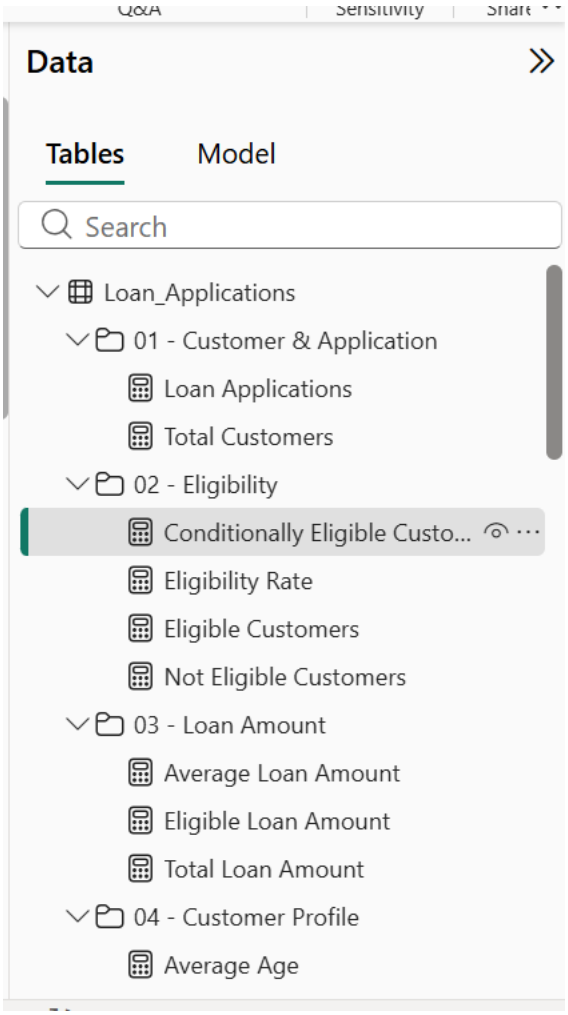


Figure 14 — *Power BI data model showing the loan application table and relationships*

3.2 DAX Measures

Six core DAX measures were created and stored in a dedicated _Measures table to maintain a clean and professional data model structure:

Measure	DAX Formula
Total Loan Applications	COUNTROWS('Loan Data')
Total Requested Loan Amount	SUM('Loan Data'[LoanAmount])
Average Requested Loan Amount	AVERAGE('Loan Data'[LoanAmount])
Eligibility Rate	DIVIDE(CALCULATE(COUNTROWS('Loan Data'), 'Loan Data'[Eligibility_Status] = "Eligible"), COUNTROWS('Loan Data'), 0) * 100
Average Policy Maximum	AVERAGE('Loan Data'[Policy_Max_Loan])
Total Customers	DISTINCTCOUNT(Loan_Applications[Applicant_ID])



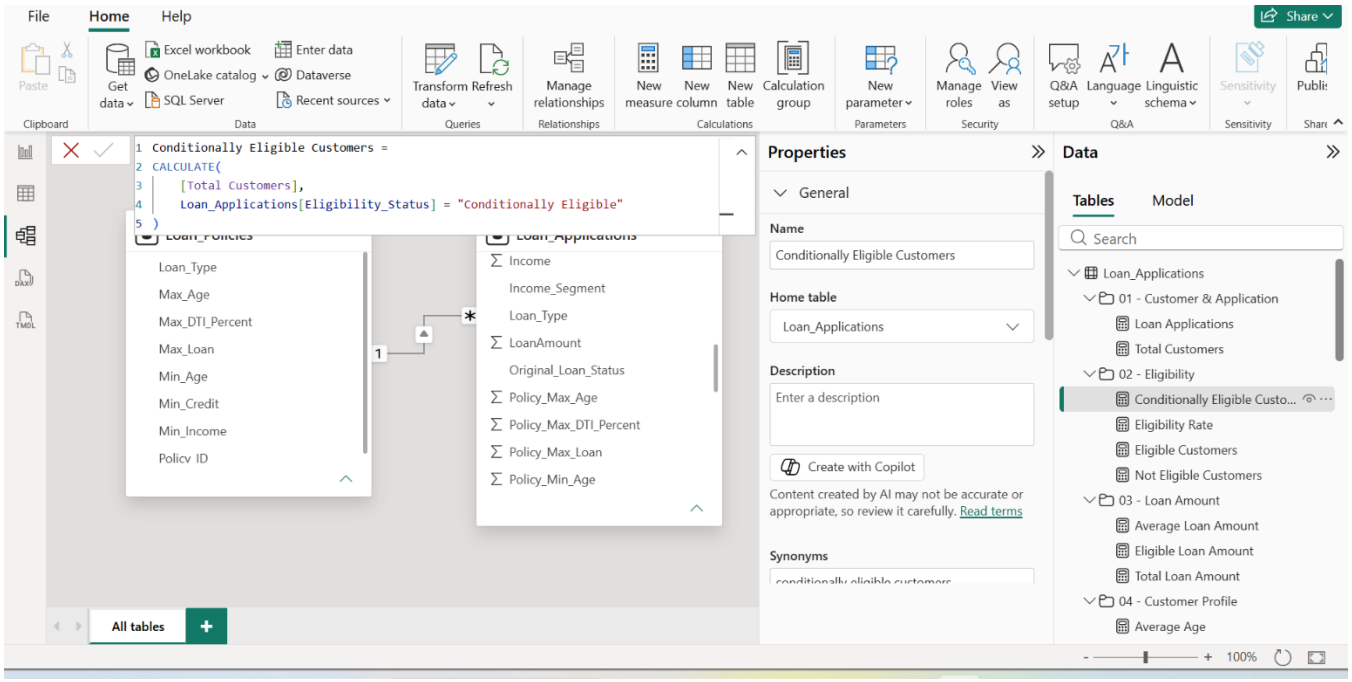


Figure 15 — *DAX measures and data model structure in Fields pane*

3.3 Page 1 — Executive Summary

The Executive Overview page summarizes key loan indicators through KPI cards for Total Customers, Eligibility Rate, High Risk Customers, Eligible Loan Amount, and Eligible Customers. It includes charts for customer eligibility distribution, applications by loan product, monthly application trends, and eligibility rates across loan categories, along with a map showing application distribution by city. Interactive filters allow users to explore customer eligibility, loan activity, trends, and regional patterns based on the selected criteria.

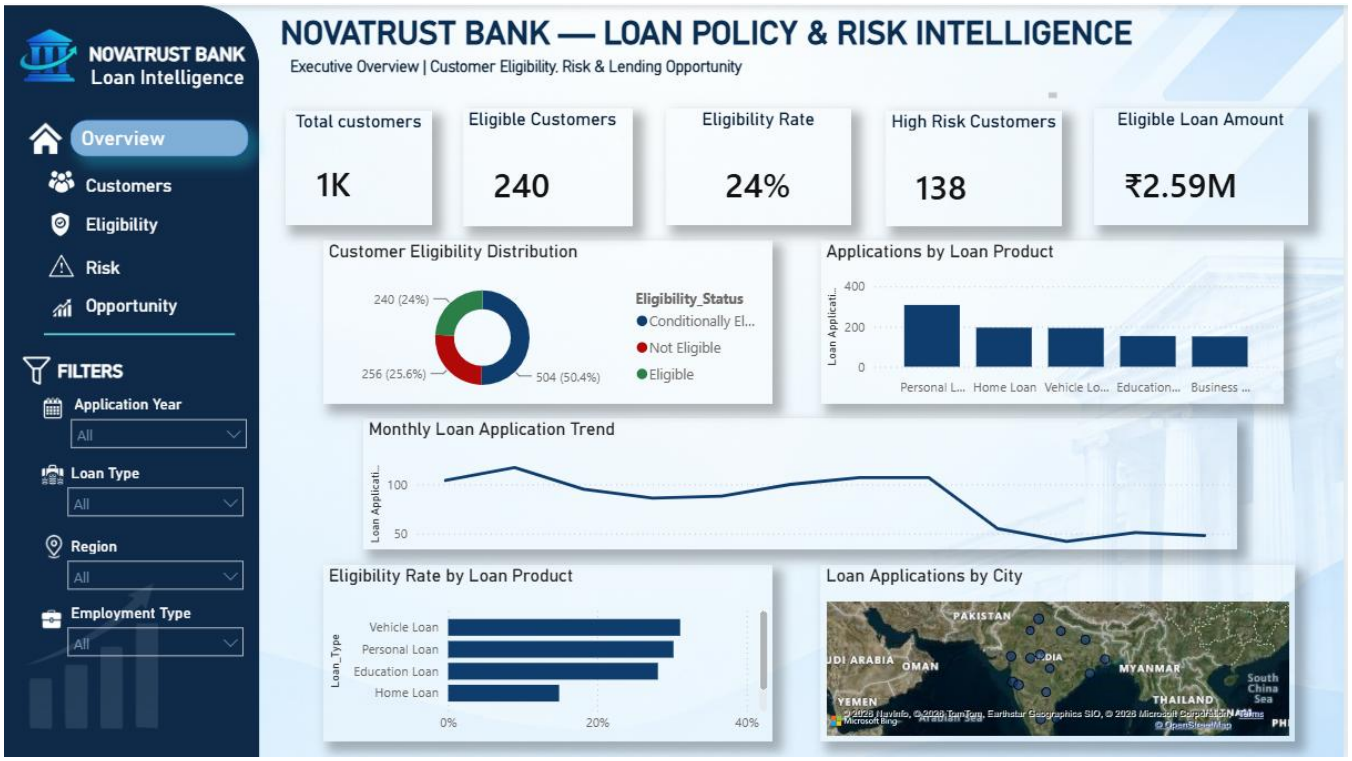


Figure 16 — *Page 1: Executive Overview dashboard*

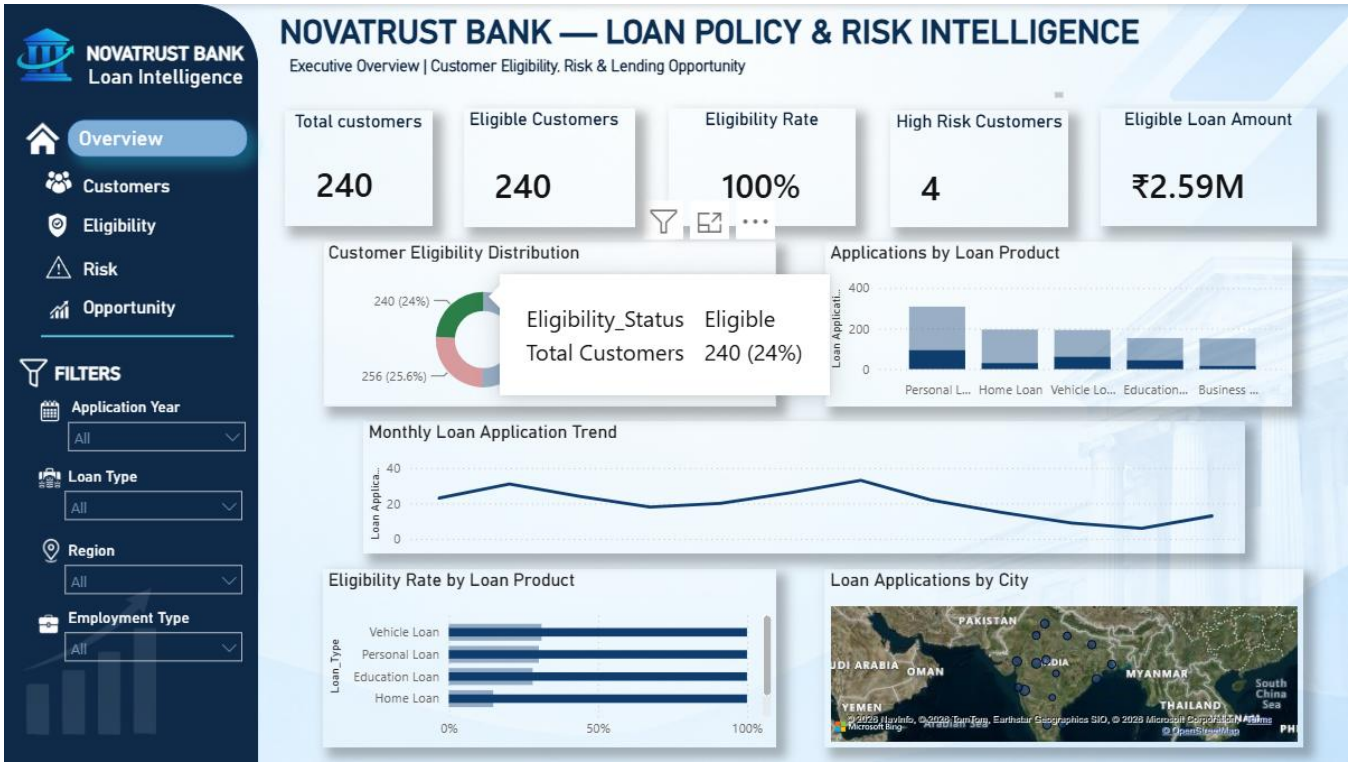


Figure 15 — *Customer Eligibility Distribution*

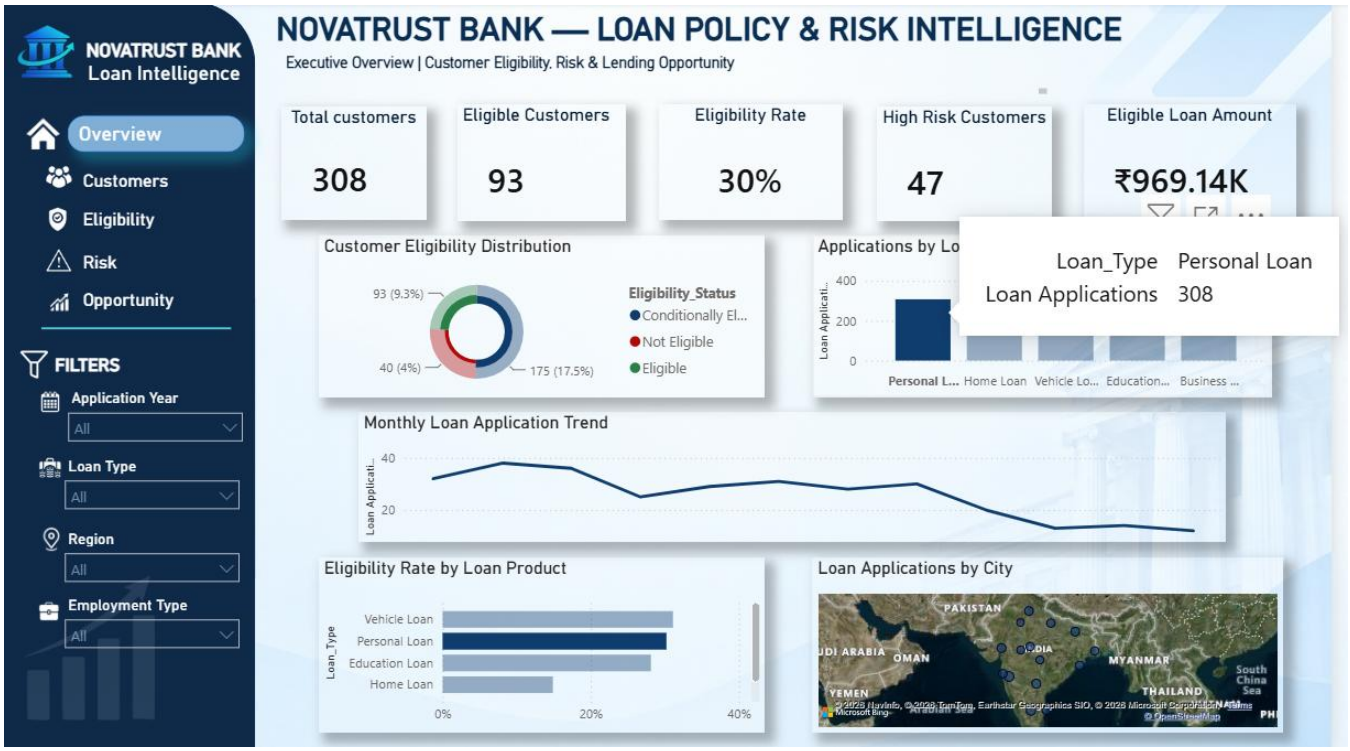


Figure 16 — *Applications by Loan Product and Monthly Loan Application Trend*

3.4 Page 2 — Customer Profile & Segmentation

The Customer Profile & Segmentation page summarizes customer demographics and financial characteristics using KPI cards for Average Age, Total Customers, Average Income, Average Credit Score, and Average DTI. It includes charts for age group distribution, income segments, and the relationship between income, credit score, and loan amount, along with a matrix comparing customer counts across income segments and credit score bands. These visuals help explore customer groups and their financial characteristics without implying causal relationships.



Income_Segment	Excellent	Fair	Good	Poor	Very Good	Total
Middle Income	88	80	44	240	52	**504**
Low Income	39	32	16	86	27	**200**
High Income	78	42	24	120	32	**296**
Total	**205**	**154**	**84**	**446**	**111**	**1000**

Figure 17 — *Page 2: Customer Profile & Segmentation dashboard*

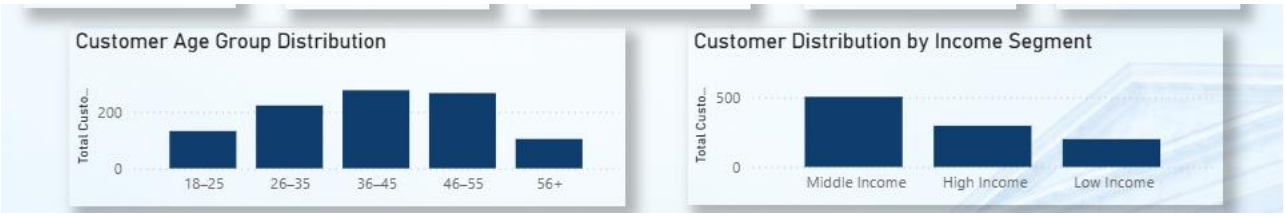


Figure 18 — *Customer Age Group Distribution and Customer Distribution by Income Segment*



Income_Segment	Excellent	Fair	Good	Poor	Very Good	Total
Middle Income	88	80	44	240	52	**504**
Low Income	39	32	16	86	27	**200**
High Income	78	42	24	120	32	**296**
Total	**205**	**154**	**84**	**446**	**111**	**1000**

Figure 19 — *Income vs Credit Score and Customer Segmentation Overview*

3.5 Page 3 — Loan Policy & Customer Eligibility

The Loan Policy & Customer Eligibility page analyzes application outcomes using KPI cards for Eligible Customers, Total Applications, Eligibility Rate, Conditionally Eligible, and Not Eligible. It includes charts showing eligibility distribution by loan type, eligibility decision reasons, requested loan amounts versus policy maximums, and eligible lending capacity by product. A summary table compares customer counts,

eligibility statuses, and average loan amounts across loan categories, supporting the review of eligibility patterns and policy-related indicators.

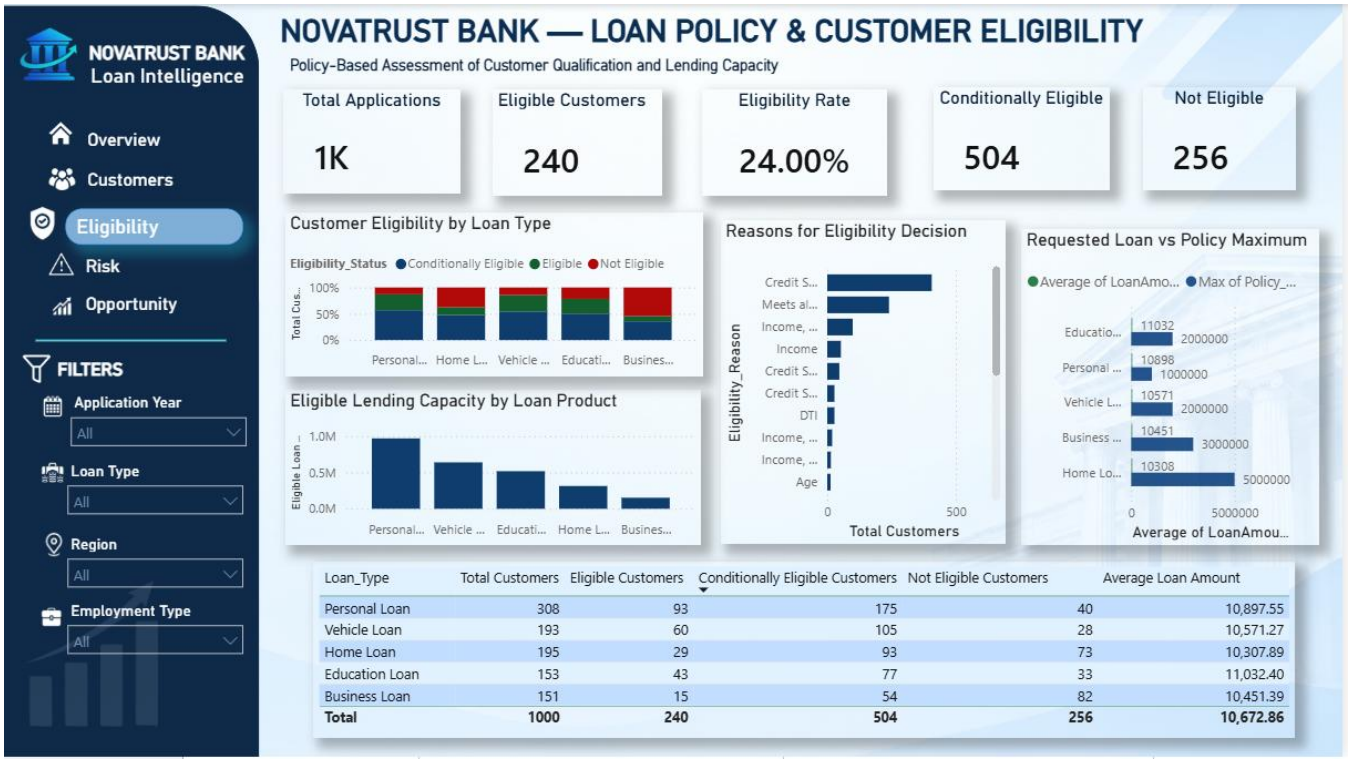


Figure 20 — *Page 3: Loan Policy & Customer Eligibility dashboard*

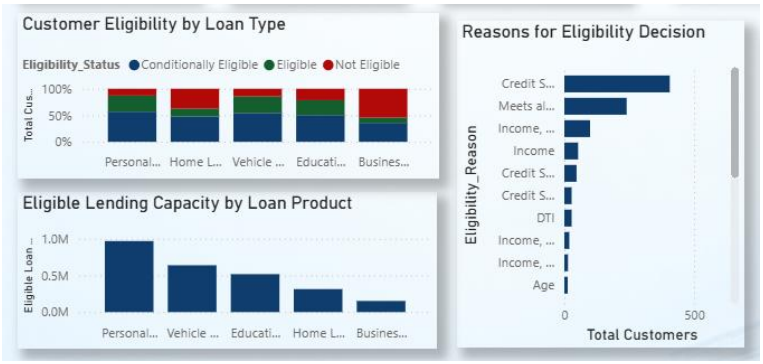


Figure 21 — *Customer Eligibility by Loan Type and Reasons for Eligibility Decision*

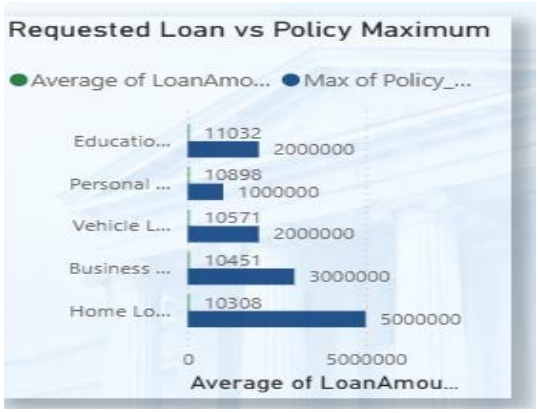


Figure 22 — *Requested Loan vs Policy Maximum*

3.6 Page 4 — Credit Risk & Loan Risk Intelligence

The Credit Risk & Loan Risk Intelligence page analyzes customer risk classifications using KPI cards for High Risk Rate, High Risk Customers, Average Credit Score, Average DTI, and Previous Defaults. It includes charts showing risk distribution by category and loan type, customer income versus credit score, and average DTI across risk groups. A detailed table summarizes loan type, risk category, applicant count, income, credit score, DTI, and previous defaults, helping identify customer groups that may require further risk assessment.

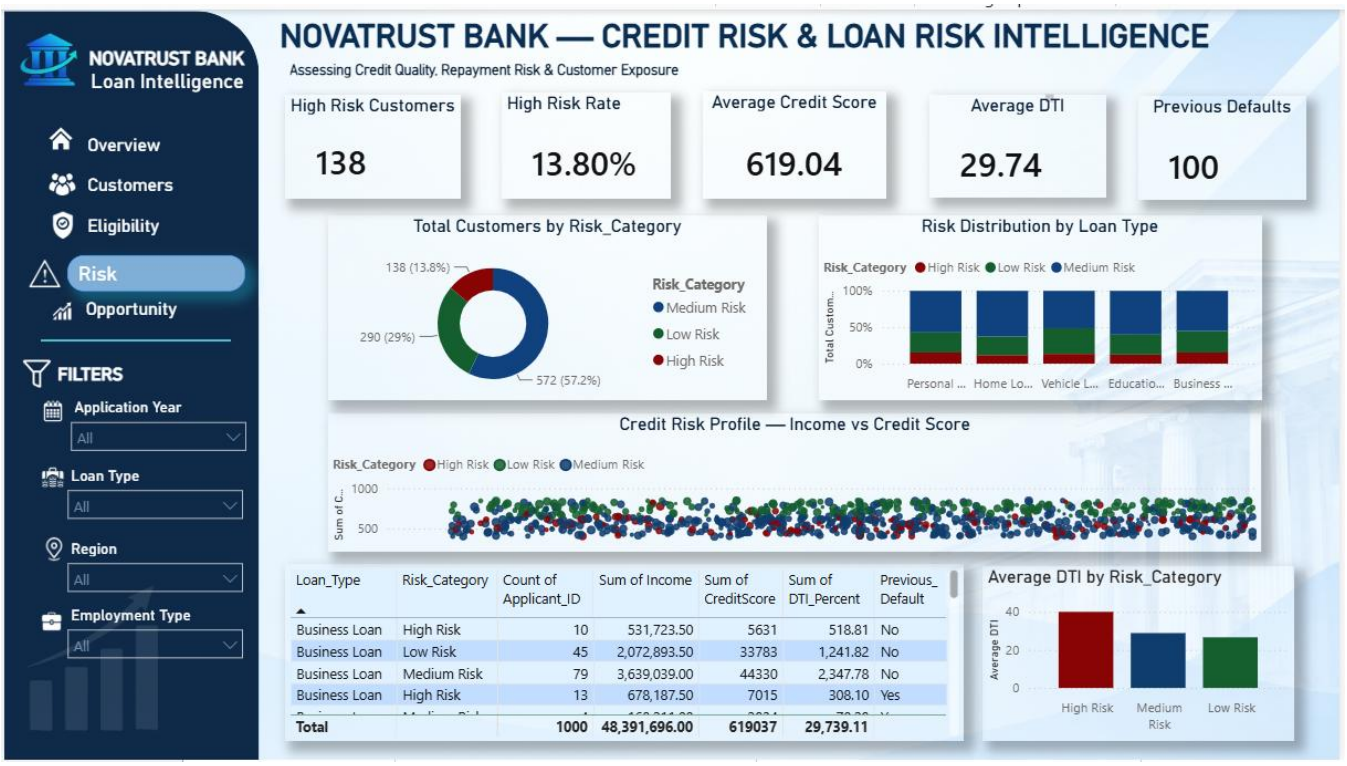


Figure 23 — *Page 4: Credit Risk & Loan Risk Intelligence dashboard*



Figure 24 — *Risk Distribution by Loan Type*

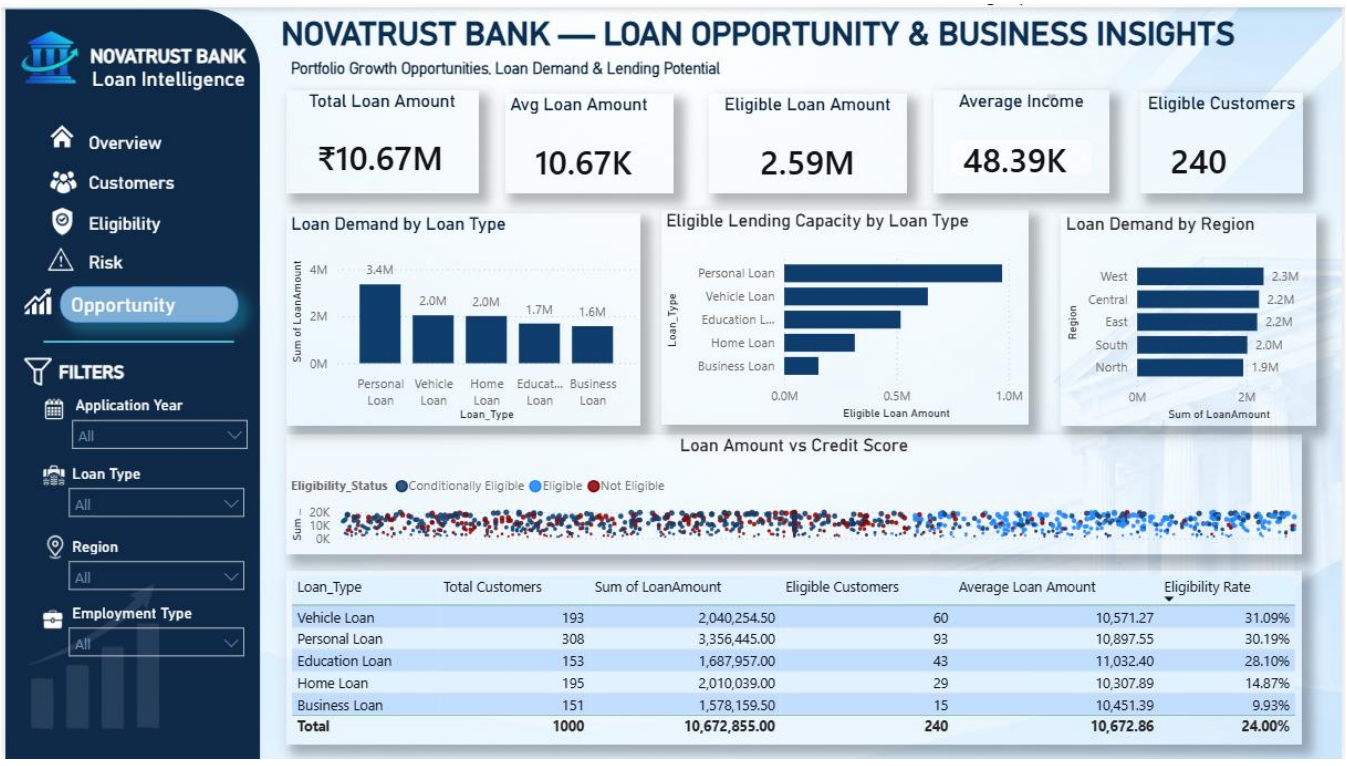


Loan_Type	Risk_Category	Count of Applicant_ID	Sum of Income	Sum of CreditScore	Sum of DTI_Percent	Previous_ Default
Business Loan	High Risk	10	531,723.50	5631	518.81	No
Business Loan	Low Risk	45	2,072,893.50	33783	1,241.82	No
Business Loan	Medium Risk	79	3,639,039.00	44330	2,347.78	No
Business Loan	High Risk	13	678,187.50	7015	308.10	Yes
Total		**1000**	**48,391,696.00**	**619037**	**29,739.11**	

Figure 25 — *Credit Risk Profile and Average DTI by Risk Category*

3.7 Page 5 — Loan Opportunity

The Loan Opportunity page analyzes loan demand and eligible lending amounts using KPI cards for Average Loan Amount, Total Loan Amount, Eligible Loan Amount, Average Income, and Eligible Customers. It includes charts comparing loan demand and eligible lending capacity by loan type, regional loan demand, and the relationship between loan amount and credit score. A detailed table summarizes loan products, customer counts, total and average loan amounts, eligible customers, and eligibility rates, helping compare lending opportunities without treating requested amounts as approved loans or confirmed profit.



Loan_Type	Total Customers	Sum of LoanAmount	Eligible Customers	Average Loan Amount	Eligibility Rate
Vehicle Loan	193	2,040,254.50	60	10,571.27	31.09%
Personal Loan	308	3,356,445.00	93	10,897.55	30.19%
Education Loan	153	1,687,957.00	43	11,032.40	28.10%
Home Loan	195	2,010,039.00	29	10,307.89	14.87%
Business Loan	151	1,578,159.50	15	10,451.39	9.93%
Total	**1000**	**10,672,855.00**	**240**	**10,672.86**	**24.00%**

Figure 26 — *Page 5: Loan Opportunity dashboard*

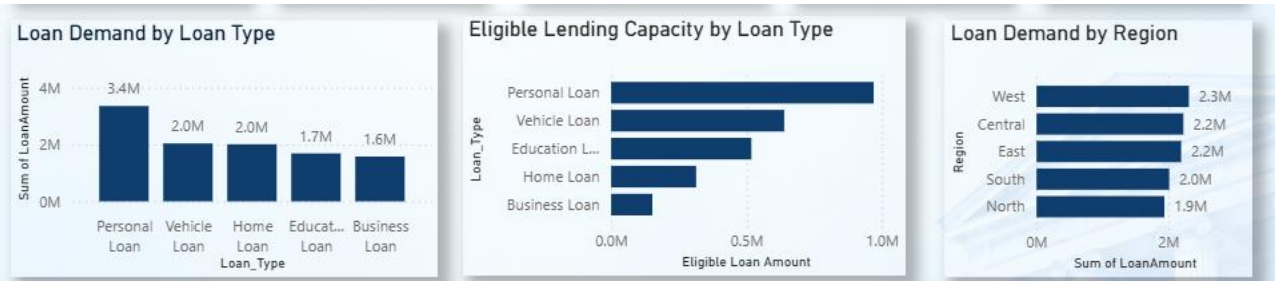
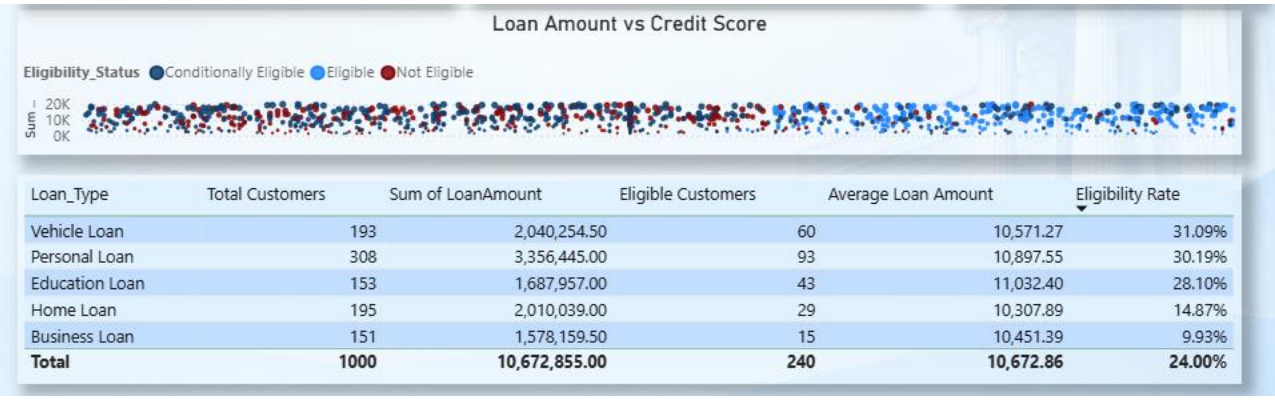


Figure 27 — *Loan Demand by Loan Type and Eligible Lending Capacity*



Loan_Type	Total Customers	Sum of LoanAmount	Eligible Customers	Average Loan Amount	Eligibility Rate
Vehicle Loan	193	2,040,254.50	60	10,571.27	31.09%
Personal Loan	308	3,356,445.00	93	10,897.55	30.19%
Education Loan	153	1,687,957.00	43	11,032.40	28.10%
Home Loan	195	2,010,039.00	29	10,307.89	14.87%
Business Loan	151	1,578,159.50	15	10,451.39	9.93%
Total	**1000**	**10,672,855.00**	**240**	**10,672.86**	**24.00%**

Figure 28 — *Loan Amount vs Credit Score*

4. Python Analysis — EDA & Time Series Analysis

4.1 Environment and Libraries

Python is used as a companion analytical tool for the banking loan dataset, with Pandas for data manipulation, data type conversion, grouping, and summary statistics, and Matplotlib for visualizing eligibility, loan demand, monthly trends, and policy maximums. This analysis complements the Power BI dashboard by supporting data review and presenting key findings through static charts, while Power BI remains the main interactive reporting tool.

```
from pathlib import Path

import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import matplotlib.dates as mdates


# ============================================================
# CONFIGURATION
# ============================================================

FILE_PATH = r"C:\Users\USER\Downloads\Banking_Loan_Policy_PowerBI_Ready.xlsx"
SHEET_NAME = "Loan_Applications"

OUTPUT_DIR = Path("python_analysis_output")
OUTPUT_DIR.mkdir(exist_ok=True)

# Consistent, professional chart formatting
plt.rcParams["figure.figsize"] = (11, 6)
plt.rcParams["font.size"] = 10
plt.rcParams["axes.titlesize"] = 13
plt.rcParams["axes.labelsize"] = 10
plt.rcParams["figure.dpi"] = 120


# ============================================================
# 4.2 DATA LOADING AND PREPROCESSING
# ============================================================

print("\n" + "=" * 65)
print("NOVATRUST BANK — PYTHON DATA ANALYSIS")
```

Figure 22 — *Python environment setup and library imports — Python IDLE*

4.2 Data Loading and Preprocessing

The Python script loads the banking dataset, standardizes column names, converts loan amounts to numeric values and Application_Date to a date-time format, and validates the required fields. It removes records with missing required values and trims whitespace from Loan_Type and Eligibility_Status to ensure consistent categories for analysis. These preprocessing steps prepare the data for visualization and are separate from transformations performed in Power Query.

```python
# 4.2 DATA LOADING AND PREPROCESSING
# ============================================================

print("\n" + "=" * 65)
print("NOVATRUST BANK — PYTHON DATA ANALYSIS")
print("=" * 65)

# Load Excel file
file_path = Path(FILE_PATH)

if not file_path.exists():
    raise FileNotFoundError(
        f"Dataset not found: {FILE_PATH}\n"
        "Place the Excel file in the same folder as this Python script."
    )

df = pd.read_excel(
    file_path,
    sheet_name=SHEET_NAME
)

# Standardize column names
df.columns = df.columns.astype(str).str.strip()

print("\nOriginal dataset loaded successfully.")
print("Original number of records:", f"{len(df):,}")
print("\nAvailable columns:")
print(df.columns.tolist())

# Required fields for the analysis
required_columns = [
    "Loan_Type",
    "LoanAmount",
    "Policy_Max_Loan",
    "Application_Date",
    "Eligibility_Status"
]

missing_columns = [
    column for column in required_columns
```

```
NOVATRUST BANK — PYTHON DATA ANALYSIS
=================================================================

Original dataset loaded successfully.
Original number of records: 1,000

Available columns:
['Applicant_ID', 'Age', 'Gender', 'Income', 'LoanAmount', 'CreditScore', 'Original_Loan_Status', 'Loan_Type', 'Employment_Type', 'Employment_Years', 'Exi
sting_Loan_Count', 'Existing_EMI', 'DTI_Percent', 'Previous_Default', 'Account_Balance', 'Region', 'City', 'Application_Date', 'Tenure_Months', 'Policy_M
in_Income', 'Policy_Min_Credit', 'Policy_Max_DTI_Percent', 'Policy_Min_Age', 'Policy_Max_Age', 'Policy_Max_Loan', 'Eligibility_Status', 'Eligibility_Reas
on', 'Risk_Category', 'Income_Segment', 'Credit_Score_Band']

-----------------------------------------------------------------
DATA PREPROCESSING SUMMARY
-----------------------------------------------------------------
Records before preprocessing : 1,000
Records retained             : 1,000
Records excluded             : 0
Columns available            : 30

Data types:
Loan_Type                     string
LoanAmount                   float64
Policy_Max_Loan                int64
Application_Date      datetime64[us]
Eligibility_Status            string
dtype: object

=================================================================
BUSINESS SUMMARY METRICS
=================================================================
Total records analyzed       : 1,000
Unique applicants            : 1,000
Total requested loan amount  : ₹10,672,855.00
Average requested loan       : ₹10,672.85
Eligible records             : 240
Overall eligibility rate     : 24.00%
Peak application month       : February 2026
Applications in peak month   : 71

Applications by loan product:
```

Figure 23 — *Python data loading and preprocessing output*

4.3 Key Business Metrics

The Python script calculates key summary indicators, including total records analyzed, unique applicants, total requested loan amount, and average loan amount. It also analyzes loan distribution by product, eligibility rates across loan types, and monthly application trends, providing supporting metrics to understand the dataset and interpret the visualizations.

Metric	Value
Total Records Analyzed	1,000 records*
Unique Applicants	1,000*
Total Requested Loan Amount	10,672,855.00
Average Requested Loan Amount	10,672.86*

```
total_records = len(analysis_df)

if "Applicant_ID" in analysis_df.columns:
    unique_applicants = analysis_df["Applicant_ID"].nunique()
else:
    unique_applicants = None

total_requested_amount = analysis_df["LoanAmount"].sum()

average_requested_amount = analysis_df["LoanAmount"].mean()

eligible_records = (
    analysis_df["Eligibility_Status"]
    .eq("Eligible")
    .sum()
)

overall_eligibility_rate = (
    eligible_records / total_records * 100
)

applications_by_product = (
    analysis_df["Loan_Type"]
    .value_counts()
    .sort_values(ascending=False)
)

eligibility_by_product = (
    analysis_df.assign(
        Is_Eligible=analysis_df["Eligibility_Status"].eq("Eligible")
    )
    .groupby("Loan_Type")
    .agg(
        Total_Applications=("Is_Eligible", "size"),
        Eligible_Applications=("Is_Eligible", "sum")
    )
```

```
==================================================================
BUSINESS SUMMARY METRICS
==================================================================
Total records analyzed       : 1,000
Unique applicants            : 1,000
Total requested loan amount  : ₹10,672,855.00
Average requested loan       : ₹10,672.85
Eligible records             : 240
Overall eligibility rate     : 24.00%
Peak application month       : February 2026
Applications in peak month   : 71
```

Figure 24 — *Python loan analysis summary output*

4.4 Loan Analysis and Visualization

A time series analysis was performed to examine the changes in Instagram content performance over time. The post_datetime field was converted to datetime format and the records were sorted chronologically. Monthly views were calculated by aggregating total views for each month, followed by the calculation of a three-month moving average to identify the underlying trend. Monthly average engagement rate was also calculated to examine changes in audience interaction over time. In addition, topic-level monthly views were analysed to compare the performance of different content topics across the study period. Finally, an Exponential Smoothing model with an additive trend was applied to the monthly views series to forecast Instagram views for the next three months.

```
# ============================================================
# CHART 1 — CUSTOMER ELIGIBILITY DISTRIBUTION
# ============================================================

eligibility_counts = (
    analysis_df["Eligibility_Status"]
    .value_counts()
)

fig, ax = plt.subplots()

eligibility_counts.plot(
    kind="bar",
    ax=ax
)

ax.set_title("Customer Eligibility Distribution")
ax.set_xlabel("Eligibility Status")
ax.set_ylabel("Number of Records")
ax.tick_params(axis="x", rotation=20)
ax.grid(axis="y", alpha=0.25)

for container in ax.containers:
    ax.bar_label(container, fmt="%.0f", padding=3)

fig.tight_layout()

fig.savefig(
    OUTPUT_DIR / "chart_1_eligibility_distribution.png",
    bbox_inches="tight"
)
```

```
# ============================================================
# CHART 2 — APPLICATIONS BY LOAN PRODUCT
# ============================================================

fig, ax = plt.subplots()

applications_by_product.plot(
    kind="bar",
    ax=ax
)

ax.set_title("Loan Applications by Product")
ax.set_xlabel("Loan Product")
ax.set_ylabel("Number of Applications")
ax.tick_params(axis="x", rotation=25)
ax.grid(axis="y", alpha=0.25)

for container in ax.containers:
    ax.bar_label(container, fmt="%.0f", padding=3)

fig.tight_layout()

fig.savefig(
    OUTPUT_DIR / "chart_2_applications_by_product.png",
    bbox_inches="tight"
)
```

```
# ============================================================
# CHART 3 — MONTHLY LOAN APPLICATION TIME SERIES
# ============================================================

fig, ax = plt.subplots()

ax.plot(
    monthly_applications.index,
    monthly_applications.values,
    marker="o",
    linewidth=2
)

ax.set_title("Monthly Loan Application Trend")
ax.set_xlabel("Application Month")
ax.set_ylabel("Number of Applications")

ax.xaxis.set_major_locator(
    mdates.AutoDateLocator()
)

ax.xaxis.set_major_formatter(
    mdates.ConciseDateFormatter(ax.xaxis.get_major_locator())
)

ax.grid(alpha=0.25)

# Annotate values when there are a manageable number of months
if len(monthly_applications) <= 24:
    for date, value in monthly_applications.items():
        ax.annotate(
            str(value),
            (date, value),
            textcoords="offset points",
            xytext=(0, 7),
            ha="center",
            fontsize=8
        )
```

```
# ============================================================
# CHART 4 — ELIGIBILITY RATE BY LOAN PRODUCT
# ============================================================

product_rates = (
    eligibility_by_product["Eligibility_Rate"]
    .sort_values(ascending=False)
)

fig, ax = plt.subplots()

product_rates.plot(
    kind="bar",
    ax=ax
)

ax.set_title("Eligibility Rate by Loan Product")
ax.set_xlabel("Loan Product")
ax.set_ylabel("Eligibility Rate")
ax.set_ylim(0, 100)

ax.yaxis.set_major_formatter(
    mtick.PercentFormatter(xmax=100)
)

ax.tick_params(axis="x", rotation=25)
ax.grid(axis="y", alpha=0.25)

for container in ax.containers:
    ax.bar_label(
        container,
        fmt="%.1f%%",
        padding=3
    )

fig.tight_layout()

fig.savefig(
    OUTPUT_DIR / "chart_4_eligibility_rate.png",
    bbox_inches="tight"
)
```

```
# ============================================================
# CHART 5 — REQUESTED LOAN VS POLICY MAXIMUM
# ============================================================

plot_data = loan_policy_comparison[
    ["Average_Requested_Loan", "Average_Policy_Maximum"]
].sort_index()

fig, ax = plt.subplots()

plot_data.plot(
    kind="bar",
    ax=ax
)

ax.set_title("Average Requested Loan vs Average Policy Maximum")
ax.set_xlabel("Loan Product")
ax.set_ylabel("Amount (₹)")
ax.tick_params(axis="x", rotation=25)

ax.yaxis.set_major_formatter(
    mtick.FuncFormatter(
        lambda value, position: f"₹{value:,.0f}"
    )
)

ax.legend(
    ["Average Requested Loan", "Average Policy Maximum"]
)

ax.grid(axis="y", alpha=0.25)

fig.tight_layout()

fig.savefig(
    OUTPUT_DIR / "chart_5_requested_loan_vs_policy_maximum.png",
    bbox_inches="tight"
)
```

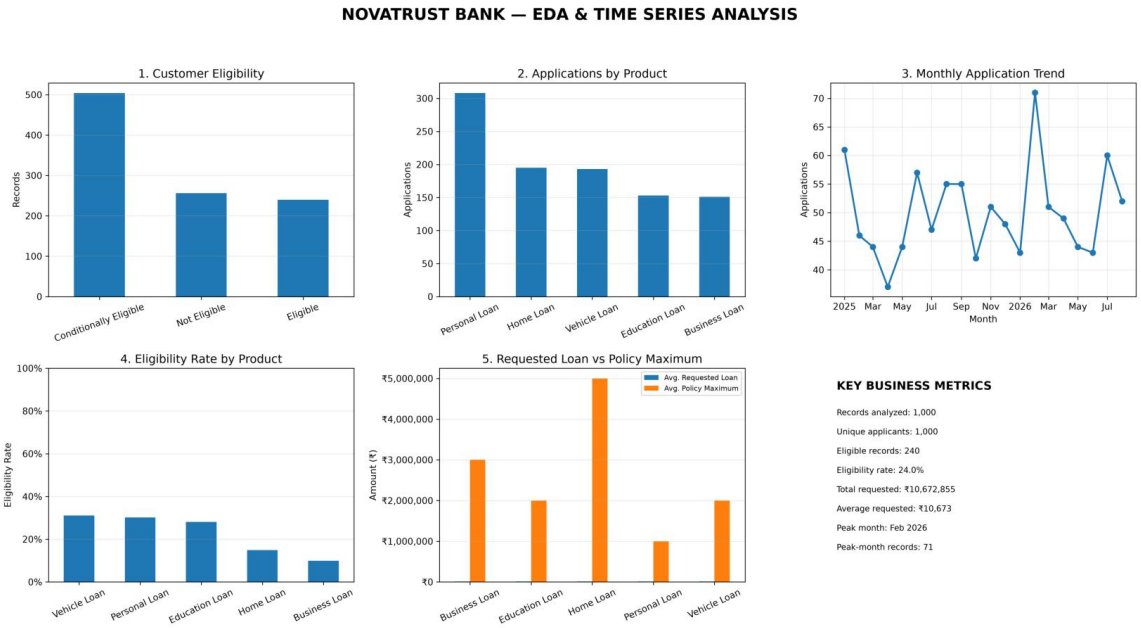


Figure 25 — *Python Time Series Dashboard — 5-chart subplot visualization*

5. Executive Recommendations

The recommendations below are based on the analysis categories implemented in the loan application project. They should be interpreted alongside the actual chart values and the bank's applicable lending rules.

Recommendation 1 — Monitor Loan Eligibility

The Customer Eligibility Distribution and Eligibility Rate by Loan Product visuals provide information about the distribution of eligibility classifications.

These indicators can be used to identify differences in recorded eligibility across loan categories and to select application groups for further review.

The eligibility definitions should remain consistent across the dataset and dashboard so that comparisons are meaningful.

Recommendation 2 — Analyse Loan Product Demand

The Applications by Loan Product and Monthly Loan Application Trend visuals show application volumes across lending categories and reporting periods.

These charts can help analysts examine which loan products account for larger application volumes and how application activity changes over time.

Reviewing these patterns can support reporting, workload planning, and further investigation of loan demand.

Recommendation 3 — Strengthen Loan Policy Monitoring

The Requested Loan vs Policy Maximum visual compares average requested loan amounts with the maximum policy limits displayed for each loan type.

Loan products where the average requested amount appears close to or above the displayed maximum may warrant further investigation at the individual application level.

Because the chart compares aggregated values, it should be used as a monitoring indicator rather than as a substitute for application-level policy checks.

6. Conclusion

The Banking – Loan Policy & Risk Intelligence project demonstrates the use of data analytics and business intelligence to examine loan application data.

Microsoft Power BI provides visual analysis of customer eligibility distribution, application counts by loan product, monthly application trends, eligibility rates, and requested loan amounts compared with policy maximums.

Power Query supports data preparation and transformation, while DAX provides measures for calculating application counts, requested loan amounts, and eligibility-related indicators. The supporting Python script uses Pandas and Matplotlib to reproduce the five principal analytical visuals and generate summary metrics.

The project organizes loan application information into a structured reporting solution that makes comparisons across loan products and eligibility categories easier to understand.

Overall, the project demonstrates how data preparation, numerical calculations, and interactive visualization can support loan policy monitoring and the analysis of lending patterns. The resulting insights can assist further investigation, while formal lending decisions remain subject to the bank's approved assessment procedures.

Appendix

A. Project File Structure

The complete project is organized in the following directory structure:

- Banking – Loan Policy & Risk Intelligence — Project folder
- Loan_Applications.csv or the actual source dataset file
- Power BI dashboard file (.pbix)
- novatrust_loan_analysis.py — Supporting Python analysis
- Final project report (.docx or .pdf)

B. Tools & Library Versions

The following tools and library versions were used in the development of this project:

Tool / Library	Version	Purpose
Microsoft Excel	Microsoft 365	Data cleaning, Dataset storage and Preparation
Power BI Desktop	2.142.1277.0 (64-bit)	Interactive dashboard, DAX and visualization
Python	3.13.5	EDA and time series analysis
Python IDLE	Python 3.13.5	Python code development and execution
Pandas	2.3.3	Data manipulation and aggregation
NumPy	1.26.4	Numerical operations and data processing
Matplotlib	3.10.8	Static data visualization
Statsmodel	0.14.5	Exponential smoothing and views forecasting
Dataset	Banking_Loan_Policy_PowerBI_Ready.xlsx	Source dataset